

Topic 2: Convexity and Gradient Descent*Lecturer: Dr. Kai Liu**Scribe: ChatGPT*

Template credit: UC Berkeley EECS; adapted from the supplied course template.

Learning goals. Recognize convex functions; connect convexity to gradients and Hessians; distinguish lower and upper curvature bounds; and prove the linear and sublinear objective-error bounds for gradient descent.

Notation. Vectors are columns, $\langle u, v \rangle = u^\top v$, and all unsubscripted vector norms in the convergence proofs are Euclidean. We write $f^* = f(x^*)$ for an attained minimum and $\Delta_k = f(x^{(k)}) - f^*$ for the objective gap.

2.1 Convex sets and convex functions

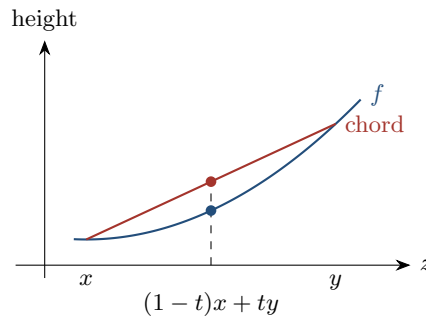
A set $C \subseteq \mathbb{R}^d$ is *convex* if $(1-t)x + ty \in C$ whenever $x, y \in C$ and $t \in [0, 1]$. Thus the entire line segment joining two feasible points remains feasible.

Definition 2.1 (Convexity) A function $f : C \rightarrow \mathbb{R}$ on a convex set C is *convex* if

$$\boxed{f((1-t)x + ty) \leq (1-t)f(x) + tf(y)} \quad (x, y \in C, t \in [0, 1]). \quad (2.1)$$

It is *strictly convex* if the inequality is strict whenever $x \neq y$ and $0 < t < 1$.

The right side is the height of the chord joining $(x, f(x))$ and $(y, f(y))$. A convex graph lies below its chords. In several dimensions, check the same property along every line segment; a function need not be increasing.



Example. For $f(x) = x^2$,

$$(1-t)x^2 + ty^2 - ((1-t)x + ty)^2 = t(1-t)(x-y)^2 \geq 0.$$

Convexity does not require differentiability: $f(x) = |x|$ is another example.

2.2 The first-order characterization

For derivative-based statements, take an open convex domain C . The gradient $\nabla f(x)$ is a column vector, so $\langle \nabla f(x), y - x \rangle$ is a scalar.

Theorem 2.2 (First-order condition for convexity) *A differentiable $f : C \rightarrow \mathbb{R}$ is convex if and only if*

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle} \quad \text{for every } x, y \in C. \quad (2.2)$$

This is a global lower bound by the tangent affine function at x . Notice the parentheses: the right side is $f(x) + \langle \nabla f(x), y - x \rangle$, not a single evaluation of f .

Proof. Convexity implies the inequality. Fix x, y and let $z_t = x + t(y - x)$, with $0 < t \leq 1$. Convexity gives

$$f(z_t) \leq (1 - t)f(x) + tf(y) \implies \frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

As $t \downarrow 0$, differentiability makes the left side converge to $\langle \nabla f(x), y - x \rangle$.

The inequality implies convexity. Let $z = (1 - t)x + ty$ with $0 < t < 1$. Apply the assumed inequality at z twice:

$$f(x) \geq f(z) + \langle \nabla f(z), x - z \rangle, \quad f(y) \geq f(z) + \langle \nabla f(z), y - z \rangle.$$

Multiply by $1 - t$ and t , respectively, and add. The gradient terms cancel because $(1 - t)(x - z) + t(y - z) = 0$. Therefore $(1 - t)f(x) + tf(y) \geq f(z)$. The endpoint cases $t = 0, 1$ hold with equality. ■

2.2.1 A concrete tangent-line calculation

For $f(x) = x^2$, the condition reads $y^2 \geq x^2 + 2x(y - x)$. Subtracting the right side leaves $(y - x)^2 \geq 0$. At $x = 1$, the tangent is $2y - 1$, and $y^2 \geq 2y - 1$ for every y .

2.2.2 Connection to minimization

If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable and convex, then

$$\boxed{x^* \text{ is a global minimizer} \iff \nabla f(x^*) = 0.} \quad (2.3)$$

Sufficiency follows immediately from (2.2). For necessity, a nonzero gradient would give a direction $-\nabla f(x^*)$ with negative directional derivative, contradicting optimality.

Scope. This zero-gradient condition is for an unconstrained minimum. For $\min_{x \geq 0} x$, the minimizer is 0 but the derivative is 1. Also, convexity alone does not guarantee that a minimum is attained: $\inf_{x \in \mathbb{R}} e^x = 0$, but e^x is never zero.

2.3 The second-order characterization: an SPSD Hessian

A symmetric matrix H is *positive semidefinite* (SPSD, or simply PSD) if $v^\top H v \geq 0$ for all vectors v . We write $H \succeq 0$. Equivalently, every eigenvalue of H is nonnegative. This does *not* mean every matrix entry is nonnegative.

Theorem 2.3 (Second-order condition for convexity) *Let f be twice continuously differentiable on an open convex set C . Then*

$$\boxed{f \text{ is convex on } C \iff \nabla^2 f(x) \succeq 0 \text{ for every } x \in C.} \quad (2.4)$$

Continuous second derivatives ensure that the Hessian is symmetric.

Proof. Necessity. For any $x \in C$ and direction v , the points $x \pm tv$ belong to C for sufficiently small $|t|$. Convexity at their midpoint gives $f(x+tv) + f(x-tv) - 2f(x) \geq 0$. Taylor expansion cancels the first-order terms:

$$f(x+tv) + f(x-tv) - 2f(x) = t^2 v^\top \nabla^2 f(x) v + o(t^2).$$

Divide by t^2 and take $t \rightarrow 0$ to get $v^\top \nabla^2 f(x) v \geq 0$.

Sufficiency. Set $h = y - x$ and integrate the second derivative along the segment from x to y :

$$f(y) - f(x) - \langle \nabla f(x), h \rangle = \int_0^1 (1-t) h^\top \nabla^2 f(x+th) h dt. \quad (2.5)$$

A PSD Hessian makes the integral nonnegative. The first-order characterization then proves convexity. ■

2.3.1 Quadratic examples

For a symmetric matrix Q ,

$$f(x) = \frac{1}{2} x^\top Q x + c^\top x + b, \quad \nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

Thus f is convex exactly when $Q \succeq 0$. For example,

$$Q = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad v^\top Q v = (v_1 - v_2)^2 \geq 0,$$

has negative off-diagonal entries but is PSD. In contrast, $Q = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has positive entries but eigenvalues 3, -1, so its quadratic is not convex.

Check the whole domain. A PSD Hessian at one point is not a global convexity test. For $f(x) = x^3$, $f''(0) = 0$, but $f''(x) = 6x$ changes sign. For $f(x) = x^4$, $f''(x) = 12x^2 \geq 0$ everywhere, so it is convex.

2.4 Examples: affine functions, norms, and squared losses

2.4.1 Affine functions

For $f(x) = a^\top x + b$,

$$f((1-t)x + ty) = (1-t)f(x) + tf(y).$$

Every affine function is both convex and concave. Its gradient is constant and its Hessian is zero.

2.4.2 Every norm is a convex function

Let N be any norm on a vector space. Positive homogeneity and the triangle inequality imply, for $0 \leq t \leq 1$,

$$\begin{aligned} N((1-t)x + ty) &\leq N((1-t)x) + N(ty) \\ &= (1-t)N(x) + tN(y). \end{aligned} \tag{2.6}$$

This covers vector norms and matrix norms, including

$$\|x\|_1 = \sum_i |x_i|, \quad \|x\|_2 = \sqrt{\sum_i x_i^2}, \quad \|x\|_\infty = \max_i |x_i|, \quad \|M\|_F = \sqrt{\sum_{i,j} M_{ij}^2}.$$

For $p \geq 1$, the ℓ_p norm is convex; in dimension at least two, the usual expression for $0 < p < 1$ is not a norm and is not convex.

Convex does not mean differentiable. The absolute value $|x|$ has no derivative at zero. For $x \neq 0$, $\nabla \|x\|_2 = x/\|x\|_2$, but this gradient is undefined at zero. Hence the smooth gradient-descent theorems later in the notes do not apply directly to a norm objective on all of $\mathbb{R}^d$.

A norm is not a strongly convex function on the whole space. For a nonzero v , a norm is affine along the ray tv , $t \geq 0$. Taking $x = v$ and $y = 2v$ gives equality in the convexity inequality. Even the Euclidean norm is not strictly convex as a function on the entire space. The term *strictly convex norm* instead refers to the geometry of its unit ball.

2.4.3 Squaring changes the curvature

The function $f(x) = \frac{1}{2} \|x\|_2^2$ has gradient x and Hessian I . It will be our simplest example with both positive lower curvature and bounded upper curvature.

More generally, $N(x)^2$ is convex for any norm N : apply (2.6), then use the fact that squaring is increasing and convex on $[0, \infty)$. It need not be differentiable or strongly convex. For instance, $\|\cdot\|_1^2$ is constant on the segment joining $(1, 0)$ and $(0, 1)$.

2.4.4 Least squares and simple closure rules

For $f(x) = \frac{1}{2} \|Ax - b\|_2^2$, $\nabla^2 f(x) = A^\top A \succeq 0$ because $v^\top A^\top A v = \|Av\|_2^2$. Therefore every least-squares loss is convex. Nonnegative weighted sums of convex functions are convex, and composing a convex function with an affine map preserves convexity. Both facts follow by applying the definition to a convex combination.

2.5 Examples: exponentials, logarithms, and maxima

2.5.1 The scalar exponential $f(x) = e^{ax}$

For any fixed $a \in \mathbb{R}$,

$$f'(x) = ae^{ax}, \quad f''(x) = a^2e^{ax} \geq 0.$$

Thus e^{ax} is convex for every a , including negative a . It is strictly convex if $a \neq 0$, and constant if $a = 0$.

For $a \neq 0$, however,

$$\inf_{x \in \mathbb{R}} a^2 e^{ax} = 0, \quad \sup_{x \in \mathbb{R}} a^2 e^{ax} = \infty.$$

Consequently there is no uniform positive lower curvature bound and no finite global upper curvature bound. It is neither globally strongly convex nor globally L -smooth on $\mathbb{R}$. Also, its infimum 0 is not attained.

On a bounded interval $[\ell, u]$, its curvature instead satisfies

$$a^2 \min\{e^{a\ell}, e^{au}\} \leq f''(x) \leq a^2 \max\{e^{a\ell}, e^{au}\}.$$

These are bounds on that interval. Using them in an algorithm requires ensuring that the relevant iterates and connecting segments stay there.

2.5.2 An exponential of an affine function

For $f(x) = \exp(a^\top x + b)$,

$$\nabla f(x) = e^{a^\top x + b} a, \quad \nabla^2 f(x) = e^{a^\top x + b} aa^\top \succeq 0.$$

Indeed $v^\top \nabla^2 f(x) v = e^{a^\top x + b} (a^\top v)^2 \geq 0$. If $d > 1$, there are directions orthogonal to a along which the function is constant, so it is not strictly convex on $\mathbb{R}^d$.

2.5.3 Two useful scalar functions

For $f(x) = -\log x$ on the convex domain $(0, \infty)$,

$$f'(x) = -1/x, \quad f''(x) = 1/x^2 > 0.$$

It is strictly convex, but has neither a finite global smoothness constant nor a positive global strong-convexity constant on that domain.

The softplus function $s(x) = \log(1 + e^x)$ satisfies

$$s'(x) = \frac{e^x}{1 + e^x}, \quad s''(x) = \frac{e^x}{(1 + e^x)^2} \in (0, 1/4].$$

It is convex and globally $1/4$ -smooth, but not strongly convex on $\mathbb{R}$, since its curvature approaches zero in the tails.

2.5.4 A nonsmooth example

For finitely many affine functions, $m(x) = \max_i (a_i^\top x + b_i)$ is convex:

$$m((1-t)x + ty) \leq (1-t) \max_i (a_i^\top x + b_i) + t \max_i (a_i^\top y + b_i).$$

A maximum can have corners; convexity does not require a Hessian to exist.

2.6 Strong convexity: a uniform lower curvature bound

Definition 2.4 (Strong convexity) For $\mu > 0$, a function $f : C \rightarrow \mathbb{R}$ on a convex set is μ -strongly convex with respect to the Euclidean norm if

$$x \mapsto f(x) - \frac{\mu}{2} \|x\|_2^2$$

is convex.

Expanding the squared norm shows that this is equivalent to

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) - \frac{\mu}{2}t(1-t)\|x - y\|_2^2. \quad (2.7)$$

The extra negative term requires a definite gap below each nontrivial chord.

2.6.1 First- and second-order forms

For differentiable f on an open convex domain, apply the first-order characterization from Section 2.2 to $f(x) - \frac{\mu}{2} \|x\|_2^2$ and simplify. The result is

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|_2^2.} \quad (2.8)$$

For twice continuously differentiable f , the corresponding condition is

$$\boxed{\nabla^2 f(x) \succeq \mu I \quad \text{for every } x.} \quad (2.9)$$

Equivalently, every directional curvature satisfies $v^\top \nabla^2 f(x) v \geq \mu \|v\|_2^2$. The *same* positive constant μ must work everywhere in the domain.

2.6.2 Consequences and distinctions

At an unconstrained minimizer, $\nabla f(x^*) = 0$, so

$$f(x) - f^* \geq \frac{\mu}{2} \|x - x^*\|_2^2. \quad (2.10)$$

Thus a minimizer, if it exists, is unique. On all of $\mathbb{R}^d$, differentiable strong convexity also ensures existence: the quadratic lower bound at 0 tends to infinity as $\|x\|_2 \rightarrow \infty$, and continuity then ensures that a minimum is attained.

Strong convexity implies strict convexity, but the converse fails. The function x^4 is strictly convex, yet its second derivative vanishes at zero, excluding every $\mu > 0$. The function e^x has positive second derivative at every point, but those values approach zero as $x \rightarrow -\infty$. Pointwise positive definiteness is weaker than a uniform bound μI .

2.6.3 Terminology used in these notes

Here, *merely convex* means that convexity is assumed without any positive strong-convexity constant. It does not exclude functions that happen to have additional curvature.

The technical term ρ -weakly convex has a different meaning: $f(x) + \frac{\rho}{2} \|x\|_2^2$ is convex. Such a function may be nonconvex; for example, $-x^2/2$ is 1-weakly convex. Our $O(1/k)$ theorem is for **convex, smooth functions**, not arbitrary weakly convex functions.

2.7 Smoothness: a uniform upper curvature bound

Definition 2.5 (*L-smoothness*) A differentiable f is L -smooth if its gradient is L -Lipschitz:

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq L \|y - x\|_2 \quad \text{for all } x, y. \quad (2.11)$$

We use $L > 0$ when choosing a step size $1/L$. An upper bound on the smallest valid smoothness constant is also valid. A larger bound produces a smaller step.

Smoothness is stronger than differentiability. Here, smoothness specifically means a Lipschitz gradient. A function can have derivatives of every order and still lack a finite global L ; e^x and x^4 are examples.

Lemma 2.6 (**Quadratic upper bound, or descent lemma**) If f is L -smooth on a convex domain, then

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|_2^2. \quad (2.12)$$

Convexity of f itself is not needed for this inequality.

Proof. Write $h = y - x$. The fundamental theorem of calculus along the segment gives

$$\begin{aligned} f(y) - f(x) - \langle \nabla f(x), h \rangle &= \int_0^1 \langle \nabla f(x + th) - \nabla f(x), h \rangle dt \\ &\leq \int_0^1 \|\nabla f(x + th) - \nabla f(x)\|_2 \|h\|_2 dt \\ &\leq \int_0^1 Lt \|h\|_2^2 dt = \frac{L}{2} \|h\|_2^2. \end{aligned}$$

The two inequalities use Cauchy–Schwarz and Lipschitz continuity. ■

2.7.1 Hessian interpretation

For a twice continuously differentiable function on an open convex domain, L -smoothness is equivalent to $\|\nabla^2 f(x)\|_{\text{op}} \leq L$ everywhere, where $\|\cdot\|_{\text{op}}$ is the spectral norm. For convex functions this becomes

$$0 \preceq \nabla^2 f(x) \preceq LI.$$

Without convexity, the spectral-norm bound instead means $-LI \preceq \nabla^2 f(x) \preceq LI$; smoothness does not imply convexity.

The two constants play different roles. For a μ -strongly convex, L -smooth function,

$$\frac{\mu}{2} \|y - x\|_2^2 \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq \frac{L}{2} \|y - x\|_2^2.$$

For a C^2 function, these assumptions are precisely $\mu I \preceq \nabla^2 f(x) \preceq LI$. Necessarily $0 < \mu \leq L$. The usual term for the upper bound is *L-smoothness*, rather than strong smoothness.

2.8 Gradient descent and the step size $1/L$

From here onward, consider unconstrained minimization on $\mathbb{R}^d$. The gradient-descent update with a fixed step size $\eta > 0$ is

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}). \quad (2.13)$$

We assume a global smoothness constant so that every step lies in a region where the required bounds hold.

2.8.1 Why this step minimizes an upper model

At a current point x , the descent lemma provides the upper model

$$m_x(z) = f(x) + \langle \nabla f(x), z - x \rangle + \frac{L}{2} \|z - x\|_2^2.$$

It is a quadratic in z with Hessian $LI \succ 0$. Its minimizer satisfies $\nabla f(x) + L(z - x) = 0$, hence

$$\arg \min_z m_x(z) = x - \frac{1}{L} \nabla f(x).$$

Thus $\eta = 1/L$ comes from minimizing a valid upper bound on the objective.

Lemma 2.7 (Decrease in one step) *Let $g = \nabla f(x)$ and $x^+ = x - \eta g$. For an L -smooth function,*

$$f(x^+) \leq f(x) - \eta \left(1 - \frac{L\eta}{2}\right) \|g\|_2^2. \quad (2.14)$$

In particular, with $\eta = 1/L$,

$$\boxed{f(x^+) \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|_2^2.} \quad (2.15)$$

Proof. Substitute $x^+ - x = -\eta g$ into (2.12):

$$f(x^+) \leq f(x) - \eta \|g\|_2^2 + \frac{L\eta^2}{2} \|g\|_2^2.$$

Collect terms. The choice $\eta = 1/L$ yields (2.15). ■

If $0 < \eta < 2/L$, the decrease is strict whenever $g \neq 0$. At $\eta = 2/L$, this bound guarantees only nonincrease. For the proofs below, use $\eta = 1/L$ throughout.

2.8.2 A scalar step-size check

For $f(x) = \frac{a}{2}x^2$ with $a > 0$, $L = a$ and

$$x^{(k+1)} = (1 - \eta a)x^{(k)}.$$

The iterates converge to zero when $0 < \eta < 2/a$. At $\eta = 1/a$, one update reaches zero. At $\eta = 2/a$, a nonzero iterate alternates signs forever, with unchanged objective value. If $\eta > 2/a$, its magnitude grows.

Why not always use step size one? For $f(x) = 50x^2$, the gradient is $100x$. A unit step multiplies the iterate by -99 , whereas the curvature-based step $1/L = 1/100$ reaches the minimizer in one step. A valid step size depends on the objective's scaling.

2.9 Linear convergence under strong convexity

Theorem 2.8 (Linear objective convergence) Suppose $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth, with $0 < \mu \leq L$. Let x^* be its unique minimizer. Gradient descent with $\eta = 1/L$ satisfies

$$\Delta_k \leq \left(1 - \frac{\mu}{L}\right)^k \Delta_0, \quad \Delta_k = f(x^{(k)}) - f^*. \quad (2.16)$$

The initial case $k = 0$ is equality; if $\mu = L$, one step reaches a minimizer.

2.9.1 Step 1: Strong convexity controls the gradient

Apply (2.8) at x , with $y = x^*$ and $g = \nabla f(x)$:

$$f^* \geq f(x) + \langle g, x^* - x \rangle + \frac{\mu}{2} \|x^* - x\|_2^2.$$

For any displacement d , completing the square gives

$$\langle g, d \rangle + \frac{\mu}{2} \|d\|_2^2 = \frac{\mu}{2} \|d + g/\mu\|_2^2 - \frac{1}{2\mu} \|g\|_2^2 \geq -\frac{1}{2\mu} \|g\|_2^2.$$

Consequently,

$$f^* \geq f(x) - \frac{1}{2\mu} \|g\|_2^2 \implies \boxed{\|\nabla f(x)\|_2^2 \geq 2\mu(f(x) - f^*)}.$$

Thus a large objective gap forces the gradient to be large. This inequality is often called a gradient-dominance or Polyak–Lojasiewicz inequality; here we have derived it from strong convexity.

2.9.2 Step 2: Combine with the smoothness decrease

Using (2.15) and the preceding gradient bound,

$$\begin{aligned} \Delta_{k+1} &\leq \Delta_k - \frac{1}{2L} \left\| \nabla f(x^{(k)}) \right\|_2^2 \\ &\leq \Delta_k - \frac{\mu}{L} \Delta_k = \left(1 - \frac{\mu}{L}\right) \Delta_k. \end{aligned}$$

Each update removes at least the fraction μ/L of the current gap.

2.9.3 Step 3: Repeat the one-step contraction

Let $q = 1 - \mu/L$. If $0 < \mu < L$, then $0 < q < 1$, and induction gives

$$\Delta_k \leq q \Delta_{k-1} \leq q^2 \Delta_{k-2} \leq \dots \leq q^k \Delta_0.$$

If $\mu = L$, Step 2 gives $0 \leq \Delta_1 \leq 0$, so the first step is optimal.

Proof in one line. Smoothness gives decrease proportional to the squared gradient; strong convexity lower-bounds that squared gradient by the objective gap. Together they give a fixed fractional decrease in the gap.

2.10 Interpreting the linear rate and the condition number

2.10.1 Linear means geometric decay

For $0 < \mu < L$, the ratio Δ_{k+1}/Δ_k is at most $q = 1 - \mu/L$ whenever $\Delta_k > 0$. This is *Q-linear* convergence of the objective gap. It is geometric (exponential) decay in k , not a straight-line decrease of $f(x^{(k)})$.

Define the condition number $\kappa = L/\mu \geq 1$. When κ is large, the update must be small enough for the steepest direction, while progress in flatter directions can be slow. Using $1 - t \leq e^{-t}$,

$$\Delta_k \leq e^{-k/\kappa} \Delta_0.$$

For $0 < \varepsilon < \Delta_0$, it is sufficient to take

$$k \geq \left\lceil \kappa \log \frac{\Delta_0}{\varepsilon} \right\rceil \quad (2.17)$$

to guarantee $\Delta_k \leq \varepsilon$. This is the usual $O(\kappa \log(1/\varepsilon))$ iteration bound when the initial gap is fixed.

2.10.2 What follows for the iterates?

Quadratic growth (2.10) gives

$$\|x^{(k)} - x^*\|_2^2 \leq \frac{2}{\mu} \Delta_k \leq \frac{2\Delta_0}{\mu} q^k.$$

Thus the distance is bounded by a constant times $q^{k/2}$. This is a geometric bound on distance; do not identify objective error and distance without the appropriate inequality.

2.10.3 A worked two-dimensional quadratic

Let

$$f(x) = \frac{1}{2}x_1^2 + 50x_2^2, \quad \nabla^2 f = \text{diag}(1, 100), \quad x^{(0)} = (1, 1)^\top.$$

Here $\mu = 1$, $L = 100$, $\eta = 0.01$, and $\kappa = 100$. The coordinates evolve independently:

$$x_1^{(k+1)} = 0.99x_1^{(k)}, \quad x_2^{(k+1)} = 0.$$

Therefore $x^* = (0, 0)$ and, for $k \geq 1$,

$$x^{(k)} = (0.99^k, 0)^\top, \quad \Delta_k = \frac{1}{2}(0.99)^{2k}.$$

The second coordinate is solved immediately, while the flatter first coordinate controls the later progress. The general bound is $\Delta_k \leq 50.5(0.99)^k$; it is valid but not exact.

Reading convergence plots. An error Cq^k becomes a line under log: $\log(Cq^k) = \log C + k \log q$. A worst-case bound describes what is guaranteed for a class of functions; a particular instance can converge faster than that bound.

2.11 Merely convex and smooth: the key one-step inequality

Now assume only that $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and L -smooth, with $L > 0$, and that a minimizer x^* exists. It need not be unique. Use $x^{(k+1)} = x^{(k)} - \nabla f(x^{(k)})/L$.

We can no longer use a positive μ to relate the gradient norm to the objective gap as in the linear proof. Instead, we track the squared distance to a fixed minimizer. The following calculation is the main step.

Lemma 2.9 (Objective gap versus decrease in squared distance) For $x^+ = x - \nabla f(x)/L$ and any minimizer x^* ,

$$f(x^+) - f^* \leq \frac{L}{2} \left(\|x - x^*\|_2^2 - \|x^+ - x^*\|_2^2 \right). \quad (2.18)$$

2.11.1 Step 1: Combine smoothness with convexity

Let $g = \nabla f(x)$. The descent lemma gives

$$f(x^+) \leq f(x) - \frac{1}{2L} \|g\|_2^2.$$

The first-order convexity condition, evaluated at x and x^* , gives

$$f^* \geq f(x) + \langle g, x^* - x \rangle \implies f(x) - f^* \leq \langle g, x - x^* \rangle.$$

Combining these two inequalities yields

$$f(x^+) - f^* \leq \langle g, x - x^* \rangle - \frac{1}{2L} \|g\|_2^2. \quad (2.19)$$

2.11.2 Step 2: Expand the squared distance after the update

Since $x^+ = x - g/L$,

$$\begin{aligned} \|x^+ - x^*\|_2^2 &= \|x - x^* - \frac{1}{L}g\|_2^2 \\ &= \|x - x^*\|_2^2 - \frac{2}{L} \langle g, x - x^* \rangle + \frac{1}{L^2} \|g\|_2^2. \end{aligned}$$

Rearranging,

$$\langle g, x - x^* \rangle - \frac{1}{2L} \|g\|_2^2 = \frac{L}{2} \left(\|x - x^*\|_2^2 - \|x^+ - x^*\|_2^2 \right).$$

Substitute this equality into (2.19) to obtain the lemma.

Why introduce distance? The two squared-distance terms will cancel across consecutive iterations. Moreover, $f(x^+) - f^* \geq 0$ implies $\|x^+ - x^*\|_2 \leq \|x - x^*\|_2$: the step does not increase the distance to any fixed minimizer.

2.12 Sublinear convergence: telescope, then use monotonicity

Theorem 2.10 ($O(1/k)$ objective convergence) *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and L -smooth, $L > 0$, and suppose its minimum is attained at x^* . Gradient descent with $\eta = 1/L$ satisfies, for $k \geq 1$,*

$$\boxed{f(x^{(k)}) - f^* \leq \frac{L \|x^{(0)} - x^*\|_2^2}{2k}}. \quad (2.20)$$

This is a bound for the last iterate, not merely an averaged iterate.

Proof. Let $D_j = \|x^{(j)} - x^*\|_2$. Apply (2.18) at iterations $j = 0, \dots, k-1$:

$$\Delta_{j+1} \leq \frac{L}{2}(D_j^2 - D_{j+1}^2).$$

Add the k inequalities:

$$\begin{aligned} \sum_{j=0}^{k-1} \Delta_{j+1} &\leq \frac{L}{2} [(D_0^2 - D_1^2) + (D_1^2 - D_2^2) + \dots + (D_{k-1}^2 - D_k^2)] \\ &= \frac{L}{2} (D_0^2 - D_k^2) \leq \frac{L}{2} D_0^2. \end{aligned}$$

Every intermediate squared distance cancels. This is the telescoping step.

By (2.15), the objective gaps are nonincreasing: $\Delta_1 \geq \Delta_2 \geq \dots \geq \Delta_k \geq 0$. Each of the k terms in the sum is therefore at least Δ_k , so

$$k\Delta_k \leq \sum_{j=0}^{k-1} \Delta_{j+1} \leq \frac{L}{2} D_0^2.$$

Dividing by k proves the result. ■

2.12.1 Iteration complexity

For a target $\varepsilon > 0$, it is sufficient to take

$$k \geq \max \left\{ 1, \left\lceil \frac{L \|x^{(0)} - x^*\|_2^2}{2\varepsilon} \right\rceil \right\}$$

to guarantee objective error at most ε . With the other quantities fixed, this is $O(1/\varepsilon)$ iterations.

2.12.2 What the theorem does and does not say

The reference sequence $1/k$ has successive ratio $k/(k+1) \rightarrow 1$; this is why its decay is called sublinear. The theorem gives an upper bound, not an assertion that every convex problem converges exactly at that speed.

For example, $f(x_1, x_2) = \frac{1}{2}x_1^2$ is convex and 1-smooth, but not strongly convex on $\mathbb{R}^2$. Its minimizers are all $(0, t)$. A step with $\eta = 1$ reaches $(0, x_2^{(0)})$, a minimizer, immediately. The $O(1/k)$ guarantee still holds; the actual behavior is faster. The proof bounds objective error but does not establish uniqueness of the minimizer or a geometric bound on distance.

2.13 Application: least squares and ridge regression

For $A \in \mathbb{R}^{n \times p}$ and $b \in \mathbb{R}^n$, consider

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2, \quad \nabla f(x) = A^\top (Ax - b), \quad \nabla^2 f = A^\top A.$$

If $A \neq 0$, a smoothness constant is $L = \lambda_{\max}(A^\top A) = \|A\|_{\text{op}}^2 > 0$. If A has full column rank, then $\mu = \lambda_{\min}(A^\top A) > 0$, so the linear theorem applies. If A is rank deficient, it is not strongly convex on $\mathbb{R}^p$, but the convex $O(1/k)$ theorem still applies because least squares attains a minimum. A singular quadratic may enjoy faster convergence on its nonzero-curvature subspace; the general convex bound need not be sharp. For $A = 0$, the loss is constant and every point is optimal.

2.13.1 Adding ridge regularization

For $\lambda > 0$, let

$$f_\lambda(x) = \frac{1}{2} \|Ax - b\|_2^2 + \frac{\lambda}{2} \|x\|_2^2.$$

Then

$$\nabla f_\lambda(x) = A^\top (Ax - b) + \lambda x, \quad \nabla^2 f_\lambda = A^\top A + \lambda I.$$

Every eigenvalue increases by λ , giving

$$\mu_\lambda = \lambda_{\min}(A^\top A) + \lambda, \quad L_\lambda = \lambda_{\max}(A^\top A) + \lambda, \quad \eta_\lambda = \frac{1}{L_\lambda}. \quad (2.21)$$

The unique minimizer solves $(A^\top A + \lambda I)x_\lambda^* = A^\top b$. All coordinates are regularized in this model, including an intercept coefficient if a column of ones is included in A .

2.13.2 A numerical conditioning example

For

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 1 & 7 & 7 \\ 1 & 8 & 9 \end{pmatrix},$$

the eigenvalues of $A^\top A$ are approximately 0.095306, 3.171707, and 297.732986. Thus the unregularized problem has condition number about 3124.

λ	μ_λ	L_λ	$\kappa_\lambda = L_\lambda/\mu_\lambda$
0	0.0953	297.733	3123.95
0.1	0.1953	297.833	1524.95
1	1.0953	298.733	272.74
10	10.0953	307.733	30.48
100	100.0953	397.733	3.97
200	200.0953	497.733	2.49

Regularization improves this condition number, but it also changes the objective and its minimizer. Faster optimization does not imply better unregularized data fit or better prediction. For a loss scaled by $1/(2n)$, divide $A^\top A$ by n before computing its curvature constants; the added λI remains unchanged.

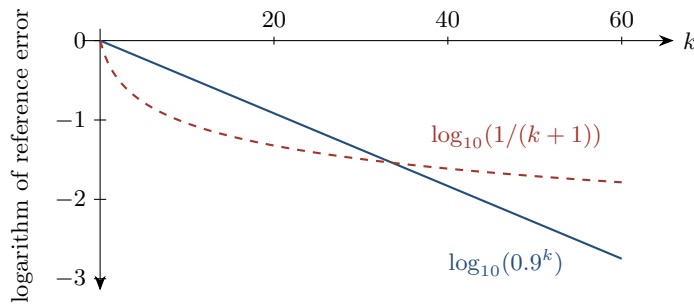
2.14 Comparing the rates and avoiding common mistakes

Both guarantees below concern ordinary gradient descent with step $1/L$ for an unconstrained objective on $\mathbb{R}^d$ with an attained minimum. Let $D_0 = \|x^{(0)} - x^*\|_2$.

Assumptions	Objective-gap guarantee	Iterations for error ε
Convex and L -smooth	$\Delta_k \leq \frac{LD_0^2}{2k}$	$O(LD_0^2/\varepsilon)$
μ -strongly convex and L -smooth	$\Delta_k \leq (1 - \mu/L)^k \Delta_0$	$O\left(\frac{L}{\mu} \log \frac{\Delta_0}{\varepsilon}\right)$

The complexity expressions describe positive initial error and the small- ε regime; if the initial point is optimal, no update is needed.

2.14.1 A picture of the two decay patterns



These are illustrative reference sequences, not measured errors from a single optimization problem. The geometric curve has constant slope on this plot; the reciprocal curve flattens as k increases.

2.14.2 Checks to make before applying a theorem

1. **Domain and existence:** Is the domain convex, and is a minimizer attained? For example, e^x on $\mathbb{R}$ has no minimizer.
2. **Regularity:** A norm is convex but generally nonsmooth. Do not apply the smooth gradient-descent proof at a nondifferentiable point.
3. **Uniform curvature:** A positive Hessian at every point need not have a uniform positive lower eigenvalue bound.
4. **Step size and scaling:** A step of one need not work. Compute L for the objective as actually written.
5. **Correct error:** Compare each regularized objective to its own minimum; objective error and parameter distance are different quantities.

2.15 Review questions and a short proof checklist

2.15.1 Questions for discussion

1. Prove convexity of $\|x\|_\infty$ directly from the definition. Where does the argument fail for an ℓ_p expression with $0 < p < 1$ in dimension at least two?
2. For $f(x) = e^{-2x}$, compute f' and f'' . Is it convex, increasing, strongly convex on $\mathbb{R}$, or globally L -smooth?
3. Give a symmetric PSD matrix with a negative off-diagonal entry. Explain why entrywise positivity is unrelated to the PSD test.
4. Show that $\frac{1}{2} \|x\|_2^2$ is 1-strongly convex and 1-smooth. What does one gradient-descent step of size one do?
5. Explain why x^4 is strictly convex but not strongly convex on $\mathbb{R}$. Is its gradient globally Lipschitz?
6. In the linear-rate proof, where is strong convexity used, and where is smoothness used? Why does setting $\mu = 0$ in that proof give no decay?
7. Derive the one-step distance inequality (2.18) without consulting the notes. Which terms cancel after summation?
8. Why is monotonicity needed to convert the sum of objective gaps into a bound for the final iterate?
9. If a problem is not strongly convex, must gradient descent be slow? Explain using the example $\frac{1}{2}x_1^2$ on $\mathbb{R}^2$.
10. For ridge regression, explain how increasing λ affects the curvature bounds, the condition number, and the minimizer.

2.15.2 Proof checklist

Linear bound. Write the smoothness decrease; derive $\|\nabla f(x)\|_2^2 \geq 2\mu(f(x) - f^*)$ by completing the square; combine them; iterate the contraction.

Sublinear bound. Combine the smoothness decrease with the convex tangent lower bound; expand $\|x^{(k+1)} - x^*\|_2^2$; sum the resulting distance differences; use $k\Delta_k \leq \Delta_1 + \dots + \Delta_k$.

Definitions to keep separate. Convexity controls chords. Strong convexity adds a uniform quadratic lower bound. Smoothness controls changes in the gradient and gives a quadratic upper bound. Neither strict convexity nor ordinary differentiability alone supplies the constants needed in the convergence theorems.

2.15.3 Further reading

The exposition and worked examples above are written for this handout. For standard treatments of the underlying results, see:

1. S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004, Sections 3.1 and 9.3. Author-hosted textbook.
2. R. Tibshirani, *Gradient Descent*, Convex Optimization 10-725/36-725, Carnegie Mellon University, Fall 2016. Lecture slides.